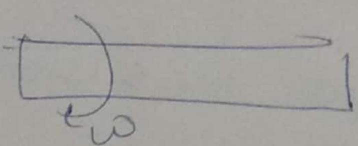
15/1 Design of shafts

Shaft : rotating machine member used for transmitting power

Axle

Spindle

from → a prime motor → apparatus



(Twisting moment + Bending)

- Dimensions → standard dimensions
- Material used
- Design consideration
  - Design for strength
  - Design for Rigidity.

Dimensions

upto 25 mm dia # 0.5 mm increment

25 mm to 50 mm dia # 1 mm increment

50 mm to 100 mm dia # 2 mm increment

100 mm to 200 mm # 3 mm increment

DB table 3.1(a) → standard dimension

Material used

The common material used for shafts are

→ Hot rolled plain C steel → extra machining

→ Cold rolled plain C steel.
  → strength is good
  → surface finish good

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Wear resistance → surface hardness → heat treatment

→ Case hardening and carburizing.

→ vaporising

→ cyaniding and Nitriding

We can also Alloy Materials.

More Heat Treatment Process

Alloys steels are used for severe applications.

It has lesser tendency to crack, wrapping and distortion.

Design considerations

→ type of load acted on shaft.

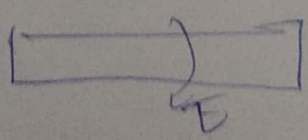
→ Bending

→ torsion

→ Axial (compression / Tension)

Ex. Shaft which is only subjected to torsion

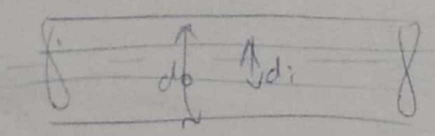
(I) subjected to Twisting moment only



Stress due to torsion

$$\tau_{xy} = \frac{16T}{\pi d_o^3 (1-k^4)}$$

$\frac{\tau}{r} = \frac{T}{J}$ DB Eq 3.1, 3.2



$r = \frac{d_o}{2}$ $J = \frac{\pi}{32}(d_o^4 - d_i^4)$

$\tau = \frac{d_o}{2} \times \frac{T}{\frac{\pi}{32}(d_o^4 - d_i^4)}$

$$\boxed{\tau_{xy} = \frac{16}{\pi}\frac{T}{d_o^3(1-k^4)}}$$

$$\boxed{k = \frac{d_i}{d_o}}$$

Q) Find the dia. of the solid steel shaft to transmit 20 kW at 200 rpm. The ultimate shear stress of the material of the shaft is 360 MPa and fos = 8

(ii) Hollow shaft with $k = 0.5$

$\tau_{xy} = \frac{16T}{\pi d_o^3(1-k^4)}$

$d_o = \left(\frac{16 T \times 8}{\pi \tau_{xy}}\right)^{1/3}$

$P = T \times \omega$

$T = \frac{P}{\omega}$

$= \frac{20 \times 10^3}{\frac{2\pi \times 200}{60}} = 954.93 Nm$

$$= \left(\frac{8}{\dots} \times 35 \times 10^{-5}\right)^{1/3}$$

$= 0.0238 \text{ km}$

~~= 23.8 mm~~

$= 47.6 \text{ mm} \Rightarrow 50 \text{ mm}$

$$\tau_{xy} = \frac{16 T \times 8}{\pi d_o^3 (1-k^4)}$$

$$d_o = \left(\frac{16 \times T \times 8}{\pi \times \tau_{xy} (1-k^4)}\right)^{1/3}$$

$= 48.63 \text{ mm} \Rightarrow 50 \text{ mm}$

$d_o = 50 \text{ mm}$

$d_i = 25 \text{ mm}$

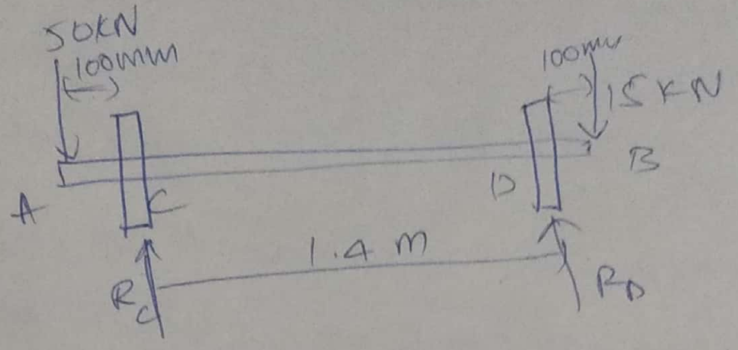
Design of shaft subjected to bending

$$\sigma_b = \frac{My}{I} \qquad y = d_o/2$$

$$I = \frac{\pi}{64}(d_o^4 - d_i^4)$$

$$\sigma_b = \frac{32 M}{\pi d_o^3 (1-k^4)}$$

Q) A pair of wheel of railway wagon carries a load of 50 kN on each axle box acting at a distance of 100 mm outside the wheel base, the gauge of rails is 1.4 m, find the dia. of axle b/w the wheels if stress is not to exceed 100MPa.



$M = 50\times10^{3}\times100\times10^{-3}$

$= 5000$ Nm.

$\sigma_b = \frac{32M}{\pi d^3} = \frac{32\times5000}{\pi\times d_o^3}$ Eq 1.1(b)

$100\times10^{6} = \frac{32\times5000}{\pi d_o^3}$

$d_o^3 = 5.09\times10^{-4}$

$d = 0.0798$ m

$\boxed{d_o = 79.8 \text{ mm}}$

$d_o = 80$ mm Table 3.5(a)

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Shaft subjected to both Bending & Torsion:

Find Equivalent Bending moment and equivalent Twisting moment.

When a shaft is subjected to both, the shaft must be designed on the basis of the two moments simultaneously.

theories (2)

(i) Max shear stress theory
(ii) Max Normal stress theory

} HW

τ - twisting moment

σ_b - Bending moment

According to Max shear stress theory,

$$\tau_{max} = \sqrt{\left(\frac{\sigma_b}{2}\right)^2 + \tau^2} \quad (1.5b)$$

~~τ_{max}~~

$$\sigma_b = \frac{32M}{\pi d^3} \quad (1.11b)$$

$$\tau = \frac{16T}{\pi d^3} \quad (1.1(d))$$

~~σ_b~~ =

$$\tau_{max} = \sqrt{\left(\frac{16M}{\pi d^3}\right)^2 + \left(\frac{16T}{\pi d^3}\right)^2}$$

$$\boxed{\tau_{max} = \frac{16}{\pi d^3}\sqrt{M^2+T^2}}$$

$\sqrt{M^2+T^2}$ - equivalent twisting moment.

hollow $d_o = \left[\frac{16}{\pi \tau_{max}}\left(\sqrt{M^2+T^2}\right) \times \frac{1}{1-k^4}\right]^{1/3}$

Eqn (3.5(b))

Acc. to max Normal stress theory

$$\sigma_{max} = \frac{\sigma_b}{2} + \sqrt{\left(\frac{\sigma_b}{2}\right)^2 + \tau^2} \qquad 1.5\ (a)$$

$$= \frac{16M}{\pi d^3} + \sqrt{\left(\frac{16M}{\pi d^3}\right)^2 + \left(\frac{16T}{\pi d^3}\right)^2}$$

$$\boxed{\sigma_{max} = \left(\frac{16M}{\pi d^3}\right)\left(1 + \sqrt{M^2+T^2}\right)}$$

$$d_o = \left[\frac{16}{\pi \sigma_{max}}\left(M + \sqrt{M^2+T^2}\right) \times \left(\frac{1}{1-k^4}\right)\right]^{1/3} \qquad 3.5(a)$$

$$\boxed{\sigma_{max} = \left(\frac{16}{\pi d^3}\right)\left[M + \sqrt{M^2+T^2}\right]}$$

Equivalent bending moment

$$M_e = \frac{1}{2}\left[M + \sqrt{M^2+T^2}\right]$$

$$M_e = \frac{\pi d^3 \sigma_{max}}{32}$$

Q/ A solid shaft subjected to bending moment of 3000 Nm and a torque of 10000 Nm the shaft is made of 45 C having ultimate tensile strength of 700 MPa and ultimate shear stress of 500 MPa assuming fos = 6, determine dia. of shaft.

Soln:

$M = 3000\ Nm$

$T = 10000\ Nm$

$\sigma_{ult} = 700\ MPa$

$\tau_{ult} = 500\ MPa$

fos = 6

$$T_e = \sqrt{M^2+T^2}$$

$$M_e = \frac{1}{2}\left(M + \sqrt{M^2+T^2}\right)$$

$$T_e = \sqrt{3000^2+10000^2} = 10440\ Nm$$

$$M_e = 6720\ Nm$$

$$\tau_{max} = \frac{16 T_e}{\pi d^3} = \frac{53170.5}{d^3}$$

$$\sigma_{max} = \frac{32 M_e}{\pi d^3} = \frac{68409}{d^3}$$

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$\sigma_{max} = \frac{68449}{d^3} = \frac{700\times10^6}{6}$

$d^3 = 5.86\times10^{-9}$

$= 0.0837$ mm

$d = 83.7$ mm

$d = 90$ mm Table(3.3(a))

$\tau_{max} = \frac{53170.5}{d^3} = \frac{500\times10^6}{6}$

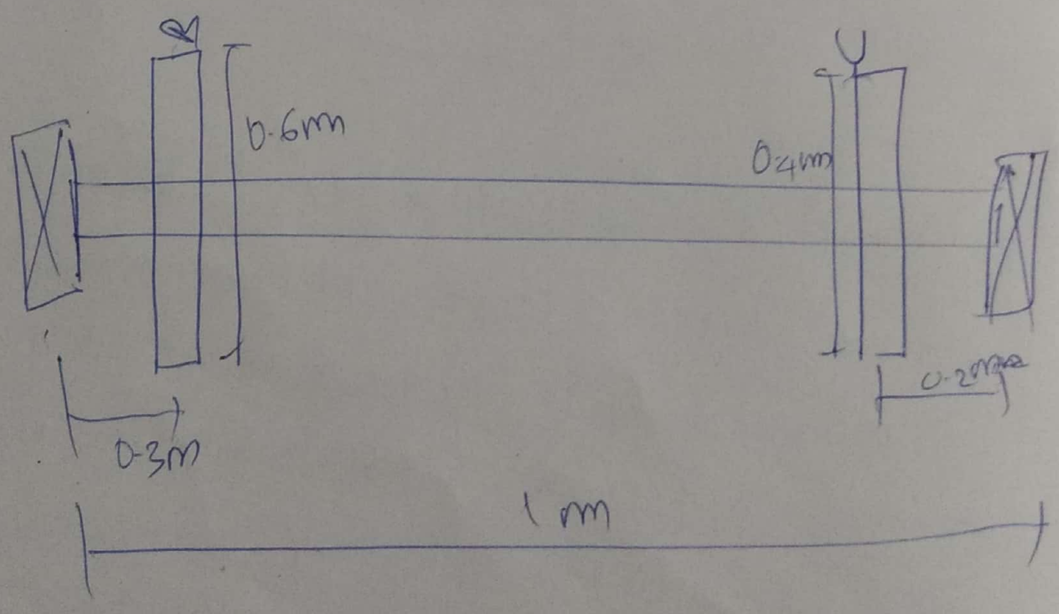
$d = 0.0860$

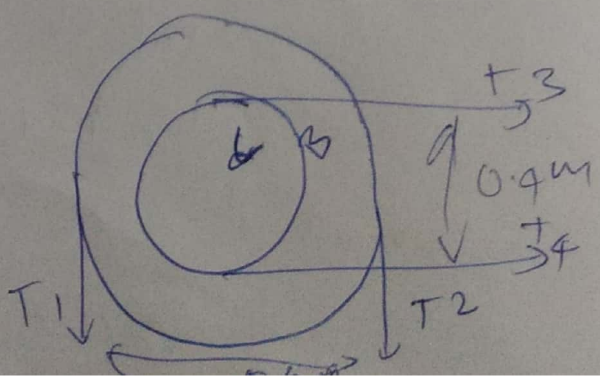
$= 86$ mm

$d = 90$ mm Table(3.5(a))

Q) A Shaft is supported by 2 bearings placed 1m apart, a 600 mm dia Pulley is mounted at a distance of 300 mm to the right of the left hand bearinging and this drives a pulley directly below it with the help of belt having max tension of 2.25 KN, Another pulley of 400 mm dia is placed 200mm to the left of the right hand bearing and

It is driven with the help of electric motor and belt which is placed horizontally to the right the angle of contact for both the pulley is 180° and $\mu = 0.24$. Determine the suitable dia for a solid shaft allowing a working of 63 MPa in tension and 42 MPa in shear for the material of the shaft. Assume that the torque of one pulley is equal to torque on another pulley.





23/1

Design of shaft based on strength.

Subjected to both bending & twisting

$$T_{eq} = \sqrt{M^2+T^2}$$

$$T_{eq} = \frac{\pi}{16} d^3 \tau$$

Step 1: Material

30C8

$\sigma_u = 400 N/mm^2$.

Step 2: Twisting moment

Q) The layout of a transmission shaft carrying 2 pulleys B and C are supported on the bearing A and D. Power is supplied to the shaft by means of a vertical belt on pulley B that is then transmitted to pulley C carrying a horizontal belt. The max tension of the belt on the pulley B is 2.5 kN. The angle of wrap is $\theta = 180°$ and the coeff. of friction is 0.24, the shaft is made of plain carbon steel 30C8 and $\sigma_{yt} = 400 N/mm^2$ and fos $n = 3$. Determine the shaft dia. based on strength basis.

Step 2: Twisting moment

Torque $T = (T_1 - T_2) R_B$

$\mu = 0.24$

$\theta = \pi$

$T_1 = 2.5\ kN = 2500N$

$\frac{T_1}{T_2} = e^{\mu\theta}$

$\frac{2500}{T_2} = e^{0.24 \times \pi}$

$\frac{2500}{T_2} = 2.125$

$T_2 = 1176.47\ N$

$$\begin{aligned} \text{Torque} &= (T_1 - T_2) R_B \\ &= (2500 - 1176.4) \times 0.25 \\ &= 330.9\ Nm \end{aligned}$$

Question (continuation)

Pulley B

dia = 500mm

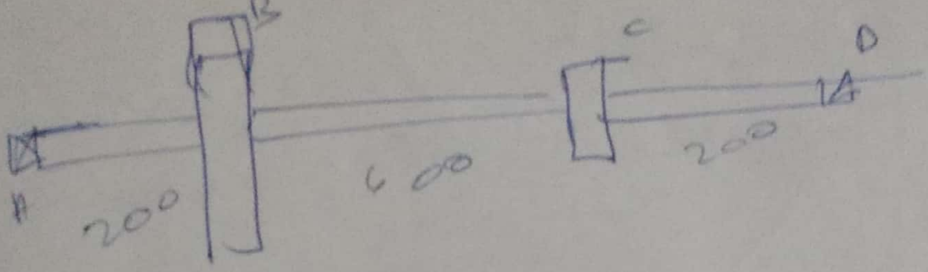
Pulley C

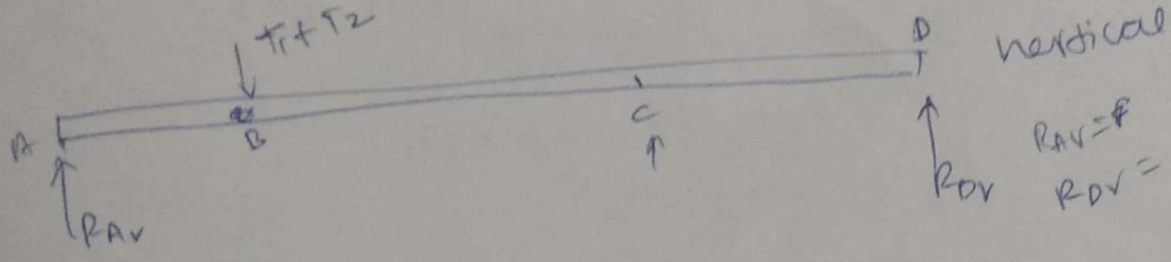
dia = 350mm

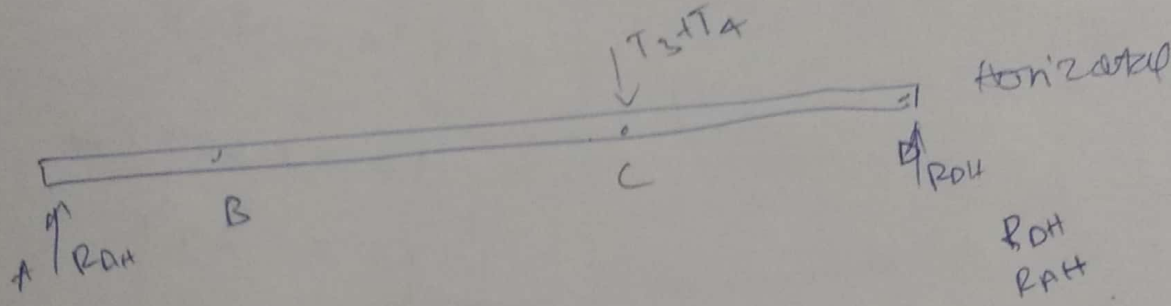
$T = 330900\ Nmm$

Step 3: Bending Moment

Find loads & reactions







vert

$T_1+T_2 = R_{AV} + R_{DV}$

$(T_1+T_2)(200) = R_{DV}(1000)$

$R_{AV} = 2941.8\ N$

$R_{DV} = \frac{T_1+T_2}{5}$

$= 735.28\ N$

Horiz

$(T_3+T_4)(800) = R_{DH}(1000)$

$R_{DH} = 0.8(\cancel{\ldots}) =$

$= \cancel{2941.12}\ N$

To find T_3+T_4 & Torque is same

$T = (T_3 - T_4) R_c$

$$\frac{T_3}{T_4} = e^{\mu\theta}$$

$330.9 = (T_3 - T_4) \times 275$

$T_3 - T_4 = 1.203$

$$\frac{T_3}{T_4} = e^{0.24 \times \pi}$$

$T_3 = 2.125 T_4$

$1.125 T_4 = 1.203$

$T_4 = 1.069$ N

$T_3 = 2.27$

$(3.34)(800) = R_{DH}(1000)$

~~$R_{DH} = 2.67$ N~~

$R_{DH} = 583$ N

$R_{AH} = 1470$ N

Vertical loading

BM @ AD = 0

BM @ B = ?

BM @ C = ?

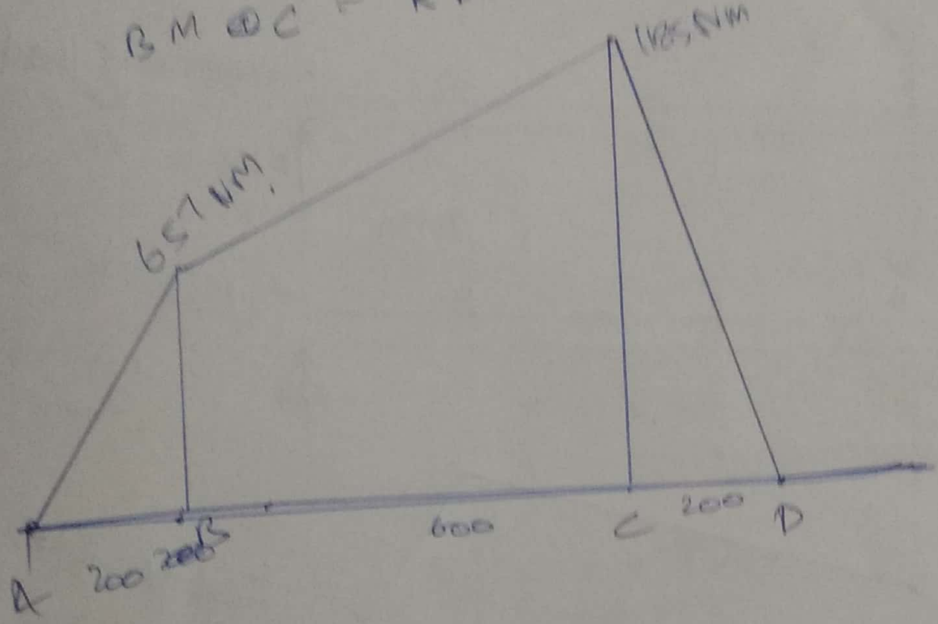
BM @B = RAV × 200 = 587.26 Nm

BM @C = RDV × 0.2 = 147.06 NM

Horizontal loading

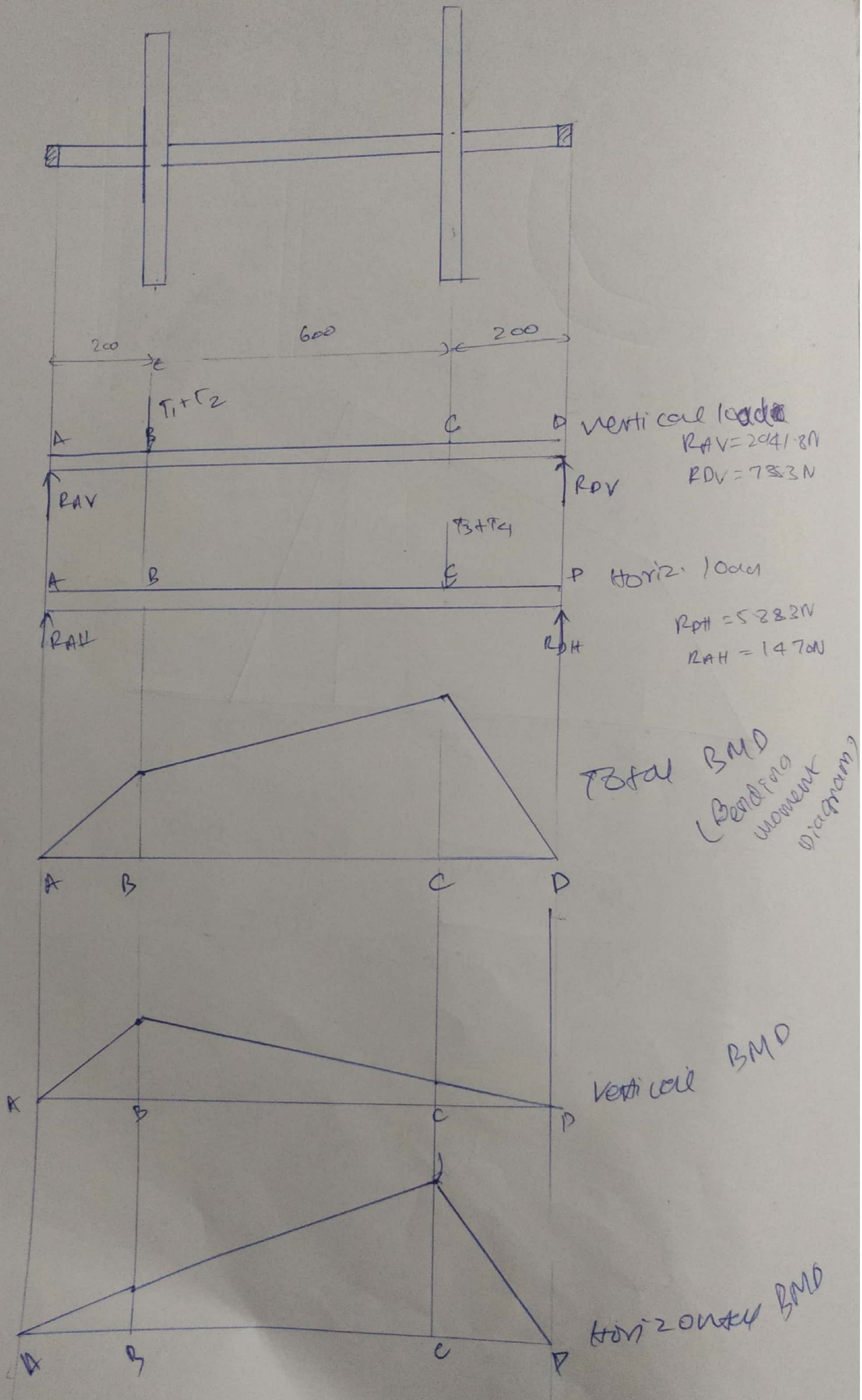
BM @ B = RAH × 0.2 = 294 NM

BM @C = RDH × 0.2 = 1176.6 NM



$$BM_B = \sqrt{BM_{BV}^2 + BM_{BH}^2} = 657.4\ Nm$$

$$BM_C = \sqrt{147^2 + 1176^2} = 1185$$


200
600
200
T_1+T_2
A
B
C
D
vertical load
RAV=2041.8N
RDV=7353N
RAV
RDV
T_3+T_4
A
B
C
D
Horiz. load
RDH=5883N
RAH=1470N
RAH
RDH
Total BMD
(Bending moment Diagram)
A
B
C
D
Vertical BMD
A
B
C
D
Horizontal BMD
A
B
C
D

Step 4

$$T_{eq} = \sqrt{M^2 + T^2}$$

$T_{eq} = \frac{\pi}{16} \tau d^3$

$\Downarrow$

1.1 (d)

$$= \sqrt{1185^2 + 331^2}$$

$$= 1230.36 \text{ Nm}$$

$$\tau = 0.5\sigma$$

shear stress at ...

$$\sigma = \frac{\sigma_u}{n}$$

$$= \frac{400 \times 10^6}{3}$$

$$= 1.33 \times 10^8 \text{ N/m}^2$$

$$\tau = 6.66 \times 10^7 \text{ N/m}^2$$

$$d^3 = \frac{16 \times T_{eq}}{\pi \tau}$$

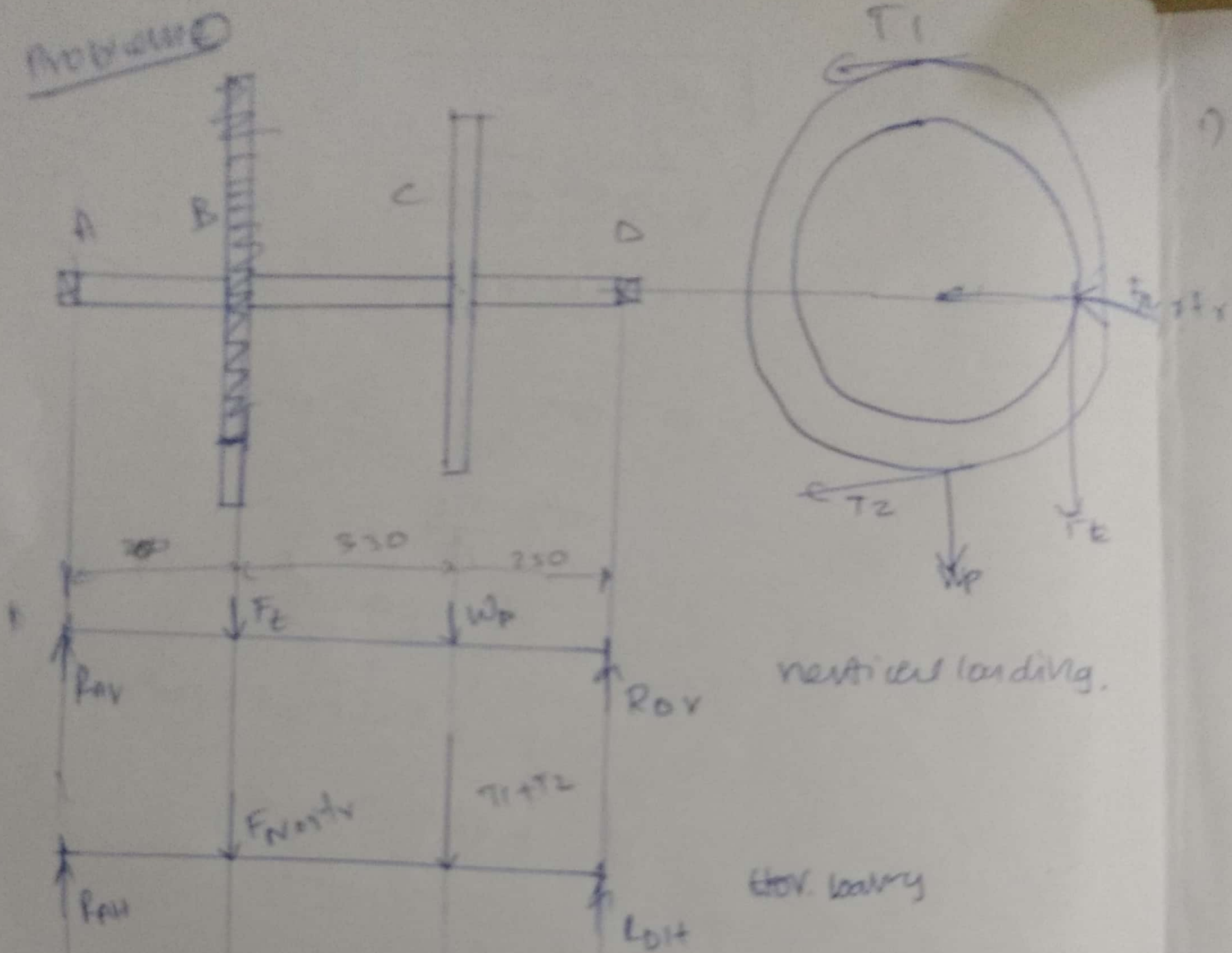
$$d = \sqrt[3]{\frac{16 \times 1230.36}{3.14 \times 6.66 \times 10^7}}$$

$$= 0.04549 \text{ m}$$

$$d = 45.49 \text{ mm}$$

50 mm Tab (8.5) (?)

Answer



T1
A
B
C
D
T2
Ft
Wp
330
250
Ft
Wp
RAV
RDV
vertical loading.
Fr
T1+T2
RAH
RDH
Hor. loading

1) Pulley $\phi = 700$ mm (drives horizontal belt)

$\theta = 180^\circ$

$W_P = 2000$ N

$t_{max}, T_1 = 3000$ N

$T_1/T_2 = 3$

$\tau = 40$ MPa

C

2) Spur gear / driver with horizontal tangential force

B

ϕ 600

$\alpha = 20^\circ$

$$T_{eq} = \frac{\pi}{16} \tau d^3$$

$$\boxed{T_{eq} = \sqrt{T^2 + M^2}}$$

$$T_1 = 3000 \text{ N}$$

$$T_2 = 1000 \text{ N}$$

$$T = (T_1 - T_2) R$$

$$= (2000)(350 \times 10^{-3})$$

$$T = 700 \text{ Nm}$$

$$T = 700 \times 10^3 \text{ Nmm}$$

$$T = f_t \times r$$

$$f_r = f_t \tan \alpha$$

$$f_t = \frac{T}{r}$$

$$= \frac{700 \times 10^3}{300}$$

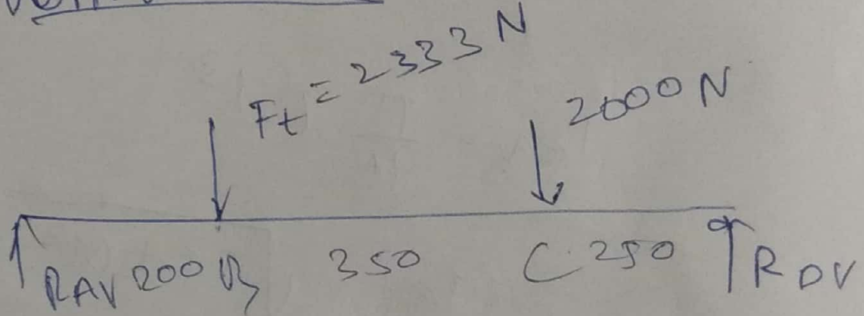
$f_t = 2333.33$ N

$f_r = 2333.33 \times \tan\alpha$

$= 849.26$ N

Bending moment

Vertical loading



$R_{AV} + R_{DV} = 4333$ N

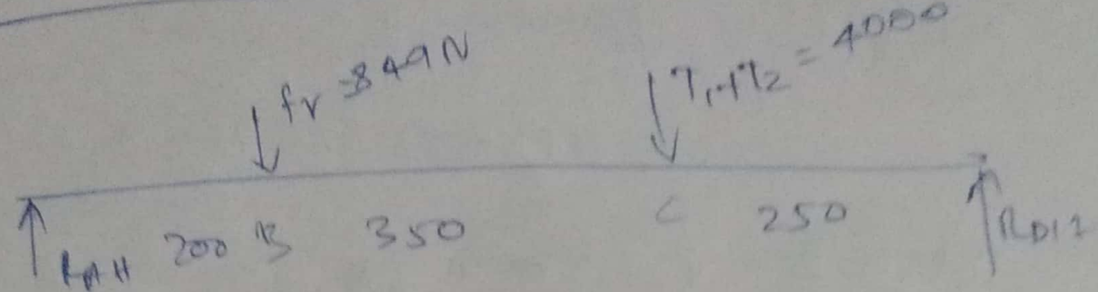
$F_t \times 200 + 2000 \times 550 - R_{DV} \times 800 = 0$

$R_{DV} = \frac{1566600}{800}$

$\boxed{R_{DV} = 1958 \text{ N}}$

$\boxed{R_{AV} = 2375 \text{ N}}$

Horizontal load



$R_{AH} + R_{DH} = 4849$

$849 \times 200 + (4000 \times 550) = R_{DH} \times 800$

$\boxed{R_{DH} = 2962\ N}$

$\boxed{R_{AH} = 1887\ N}$

BM@A, D = 0

Vert. BM@B = $\times 200$ $= R_{AV} \times 200$

= 466600 N mm $= 475000$

BM@C = 2000×250 $= 489500$

= 500000 N mm

Hor BM@B = 849×200 $= 377400$

= 169800 N mm

BM@C = 4000×250 $= 740500$

= 1000000 N mm

Total BM@B = $\sqrt{466600^2 + 169800^2}$

= 496535 N/mm

$BM@C = \sqrt{500000^2 + 1000000^2}$

$= 1118033.98 \ Nmm$

Total $BM@B = \sqrt{475^2 + 377.4^2} = 606.67$

$BM@C = \sqrt{\ } = 887.66 \ Nm$

$T_{eq} = \sqrt{T^2 + M^2}$

$= \sqrt{700^2 + \cancel{1118034}^2}$ 887.66^2

$= \cancel{1314040} \ Nmm$

$= 1130.46$

$T_{eq} = \frac{\pi}{16} \tau d^3$

$1130 = \frac{\pi}{16} \times 40 \times 10^6 \times d^3$

$d = 0.05517 \ m$

$d = 55.17 \ mm$

$d = 56 \ mm$

$d = 52.39$ mm

$\boxed{d = 56 \text{ mm}}$